

A self-learning system is one that can learn naturally. This is where the first big advantage of an open system kicks in. An open system, by virtue of its very nature, has a larger community than a closed system. As a direct result of this, it has a far larger amount of information at any point in time than its closed counterpart. And this information doubles, triples or quadruples much faster in an open system than it can in a closed system. The reasons aren't hard to find. The community that supports an open system comprises a dedicated bunch of people who want the system to develop, and this community expands every day.

So when a system has a lot of information, what does it do? In fact, what does everybody do with a lot of information? Simple. They use it. Similarly, so does the system. A lot of information means that you can single out information that makes sense, and use it to make some more sense out of it.

This is what a system can do when it learns naturally. Let us consider a hypothetical example. Assume we have a system that has some complete sentences like, "Sachin plays snooker," "Rahul plays the drums," "Rita plays the sitar," etc. Now, owing to some problems with the medium of transmitting information into the system, some other incomplete sentences have entered. So we have some discrete and seemingly random words like 'Sumesh', 'guitars', 'Rishi', 'hockey'. As of now, the complete sentences make sense, but the words make no sense at all. The system, however, notices that in the complete sentences, a proper noun is related to a common noun by the verb 'plays'. This is the information it has gathered from the information it has. If the system is even more primitive, with no inkling about what a proper noun or a common noun is, we can assume that it just observes that 'a word-that-begins-with-an-upper-case' 'plays' 'a-word-that-begins-with-a-lower-case'. So, using this newly gathered information, it can deduce that there is a likelihood that "Sumesh plays guitars" and "Rishi plays hockey," or "Sumesh plays hockey" and "Rishi plays guitars."

So now, the random words make more sense than they did initially. Thus, the system has learned. Of course, this is a very simple example, but it goes to show how a system can learn from the information it has, to make more sense out of it, and use this new 'sense' to correlate things. This, in fact, forms the basis of the two huge areas in computer science: data mining and machine learning—the ways of making sense out of information, and the ways of making a system learn or correlate. Coming back to an open system, which has a lot more information than a closed system, there is more opportunity for it to learn, and to make more sense than a system that is closed. Therefore, it evolves faster.

Thus, we have seen how an open system evolves. Since it is dynamic, it keeps evolving, learns from its evolution and evolves even further. And in order to make it evolve naturally, we must not place any artificial limitations on its ability to evolve. This can be done only if the information that a system has can flow equally to all parts of the system. If there is a concentration of information in any one part, it hampers the process of natural evolution and brings about differences within the system. And

this is bad, because quite simply, such a system cannot stand the test of time. Too many differences within a system and ... poof! It implodes.

Let us now travel back in time to 1984 to when Sun Microsystems was but a two-year-old firm. It was the year that John Gage (who, incidentally, was the 21st employee of the firm) famously said, "The network is the computer," —one line that would shape the future of computer networks, the Internet, which was still in its infancy, and the Web, which was still non-existent. These words heralded the era of networking and distributed computing. Today, at a time when cloud computing is a term that is overused to the point of becoming hackneyed, such sentences make us think and wonder.

Indeed, the level to which the Internet has grown today is probably far beyond the wildest dreams of John Gage back in 1984. But the good thing is, we do have a developed Internet today. And no, it's not just a net of inter-connected devices; we have ways of accessing information on the Internet too. This is fundamentally what the Web is—the Internet and its users accessing information on it; then going on to use the information and make the Internet grow even further.

Now, coming back to our discussion on open systems, the Web is an open system. Anyone can enter the Web, add information to it and harness other information for their benefit. The millions of people who are connected by the Internet form the community that is responsible for its existence. All the information that exists on the Internet is courtesy its users, who are the same people who use the Internet to get information as and when they want. The users trust the Internet, and therefore they want it to thrive. And as more and more people enter the Web every day, there is more and more information flowing into the Internet—making the Web learn and evolve faster.

Thus, the need of the hour is to maintain the Internet. We, the users, and the contributors of the Web, are the ones who are responsible for its sustenance. And to allow the Web to survive, we need to ensure that it continues to evolve naturally. To do this, we must keep the Web open. There shouldn't be any restrictions on anyone's ability to do anything on it. All should have the freedom to do whatever they want on the Web. If what they do is wrong, adversely affecting the Web, the community will automatically respond to make sure that the effect of the wrongdoing is nullified—the same tried-and-tested method that has worked wonders for Wikipedia. This is the broad idea of any open system—the community matters, not the individual. Individual activities and opinions do not matter much in the face of what the community does. It is the general interest in the system that overrules any individual intent or purpose, and nullifies it. Science-fiction enthusiasts will be able to correlate this concept with Hari Seldon's ideas about psychohistory in Asimov's epic Foundation series.

So, here's the big question: How do we keep the Web open? To answer this, let us again enlist the three basic features of an open Web. One, everybody takes part. Two, information never stagnates. Three, there are no restrictions whatsoever. The first point is taken care of, limited only by the availability and cost of the Internet in